

Name: _____ Reg #: _____ Section: _____

National University of Computer and Emerging Sciences, Lahore Campus

Course: Artificial Intelligence

Course Code: AI2002

Program: BS(Computer Science)

Semester: Spring 2026

Section: BCS-6B

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Exam: Quiz 1

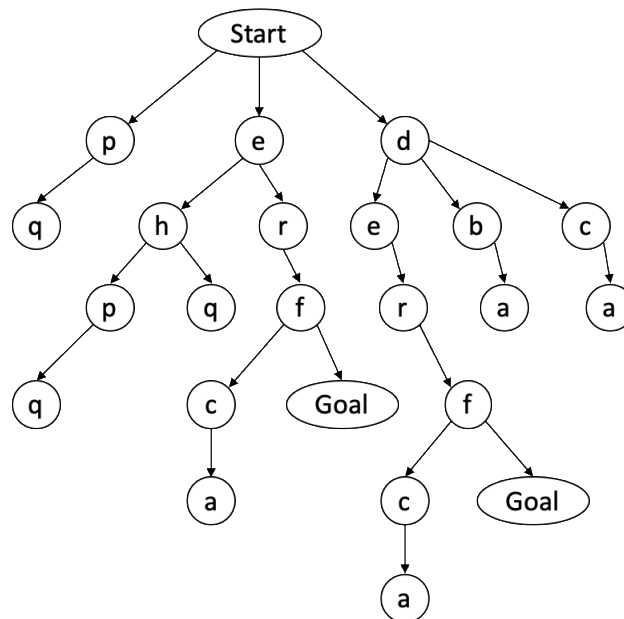
Q1) Analyze task environments based for each of the given agent based system.

(3)

Task Environment	Discrete/Continuous	Fully/Partially Observable	Deterministic/Stochastic	Sequential/Episodic	Single/Multiagent	Static/Dynamic
GPS System						
Voice Activated Mobile Assistant						

Q1) Consider the following state space, where each node represents a state, and each directed link represents an action.

(7)



Suppose the state space is to be searched, from “Start” state to “Goal” state. List the nodes of the state space in the order that they are expanded (open list) and visited (closed list) using iterative deepening search. (0.5+1+1.5+2+2)

Iteration	Open	Closed
1	[Start] []	[] [Start]
2	[Start]	[]

	[p, e, d] [e, d] [d] []	[Start] [p, Start] [e, p, Start] [d, e, p, Start]
3	[Start] [p, e, d] [q, e, d] [e, d] [h, r, d] [r, d] [d] [e, b, c] [b, c] [c] []	[] [Start] [p, Start] [q, p, Start] [e, q, p, Start] [h, e, q, p, Start] [r, h, e, q, p, Start] [d, r, h, e, q, p, Start] [e, d, r, h, e, q, p, Start] [b, e, d, r, h, e, q, p, Start] [c, b, e, d, r, h, e, q, p, Start]
4	[Start] [p, e, d] [q, e, d] [e, d] [h, r, d] [r, d] [f, d] [d] [e, b, c] [b, c] [a, c] [c] []	[] [Start] [p, Start] [q, p, Start] [e, q, p, Start] [h, e, q, p, Start] [r, h, e, q, p, Start] [f, r, h, e, q, p, Start] [d, f, r, h, e, q, p, Start] [e, d, f, r, h, e, q, p, Start] [b, e, d, f, r, h, e, q, p, Start] [a, b, e, d, f, r, h, e, q, p, Start] [c, a, b, e, d, f, r, h, e, q, p, Start]
5	[Start] [p, e, d] [q, e, d] [e, d] [h, r, d] [r, d] [f, d] [c, Goal, d] [a, Goal, d] [Goal, d]	[] [Start] [p, Start] [q, p, Start] [e, q, p, Start] [h, e, q, p, Start] [r, h, e, q, p, Start] [f, r, h, e, q, p, Start] [c, f, r, h, e, q, p, Start] [a, c, f, r, h, e, q, p, Start]